

Sirada Kittipaisarnkul

+31616336604 — p.siradakit@gmail.com — linkedin.com/in/sirada-kittipaisarnkul-068854224/ — github.com/99ffx

Summary — Engineer with a Bachelor’s in Electronics and Telecommunications Engineering and a Master’s in Medical Imaging and Applications, bringing a solid foundation in data communication, signal processing, and radio communication systems. Experienced in software development and network-related projects, combining programming skills with system integration knowledge. Strong in cross-disciplinary problem-solving, having applied AI, image processing, and signal analysis across healthcare and engineering domains. Adept at working in multinational teams, with a track record of delivering technical solutions in research and industry settings.

Skills

AI/ML Deep Learning, Machine Learning, Image Processing, Computer-Aided Diagnosis	Frontend React, Angular, HTML, CSS
Programming Python, JavaScript, TypeScript, PyTorch, NumPy, OpenCV	Workflow Agile (Scrum, SAgE), Git, Docker
	Languages Thai (Native), English (Fluent)

Experience

Radboud University Medical Center, The Netherlands Feb 2025 – July 2025
Research Intern - Automated Detection of DRESS Syndrome

- Conducted research in histopathology-based disease classification using whole-slide imaging and AI techniques.
- Contributed to the design and development of deep learning methods in collaboration with multidisciplinary teams.
- Supported dermatopathology research with findings relevant for potential clinical applications.

Danube Private University, Austria July 2024 – Nov 2024
Research Intern – Reproducibility Analysis of MRI Radiomics Features Using 3D-Printed Silicone Phantoms

- Performed segmentation and radiomics feature extraction on CT and MRI scans using **3D Slicer**.
- Processed **DICOM** images of silicone phantoms simulating soft tissue imaging properties.
- Assessed reproducibility of radiomics features across imaging modalities and acquisition parameters.

Allianz Technology Thailand, Thailand July 2022 – Aug 2023
Front End Developer

- Designed and maintained **responsive web application interfaces** using the **Angular framework**, improving user experience and interface performance.
- Collaborated within an **Agile/Scrum** environment, participating in sprints, daily stand-ups, and iterative development cycles to ensure timely delivery of high-quality features.
- Gathered and analyzed requirements with clients and stakeholders, aligning technical development with business objectives and user expectations.
- Enhanced cross-functional team communication to bridge technical and non-technical perspectives, ensuring smooth project execution.

Securities and Exchange Commission of Thailand, Thailand June 2021 – Aug 2021
Cyber Security Intern

- Monitored and analyzed **network traffic** to identify and filter **malicious IP addresses**, safeguarding network integrity and preventing cyber threats.
- Identified, documented, and reported **security risks** and incidents, ensuring rapid response and remediation.
- Developed and presented **vulnerability assessment reports**, providing actionable recommendations for mitigation and security protocol enhancement.
- Contributed to strengthening the organization’s cyber defense posture by applying threat analysis and risk management techniques.

Education

Erasmus Mundus Joint Master’s Degree in Medical Imaging and Applications (MAIA) Sept 2023 – July 2025
Université de Bourgogne (France), Università degli Studi di Cassino e del Lazio Meridionale (Italy), University of Girona (Spain)

- International 2-year, 120 ECTS program combining medical imaging, image processing, and AI-driven computer-aided diagnosis (CAD).
- Studied across three European universities: foundational imaging and acquisition (France), advanced computer vision and machine learning (Italy), applied CAD and medical applications (Spain).

Bachelor of Engineering July 2018 – July 2022
Electronics and Telecommunication Engineering King Mongkut’s University of Technology Thonburi

- Specialized in data communication, signal processing, and radio communication systems.
- Strong foundation in electronics, including microprocessors and embedded systems.
- Gained practical experience through laboratory work and research in telecommunication system design and performance analysis.

Projects

Automated Detection of DRESS Syndrome

Feb 2025 – July 2025

- Developed a weakly supervised deep learning pipeline for classifying DRESS syndrome versus Morbilliform Drug Eruption (MDE) from skin biopsy whole-slide images.
- Leveraged Multiple Instance Learning (MIL) with features from multiple foundation encoders, achieving 94.8% accuracy and 90.9% F1-score.
- Created attention-based interpretability maps to identify diagnostically relevant histological patterns, supporting potential clinical integration in dermatopathology.

Brain Tissue Segmentation

Oct 2024 – Jan 2025

- Segmented brain tissues (CSF, WM, GM) from IBSR1B using preprocessing, bias correction, histogram matching, and multi-atlas segmentation with voting and fusion. Applied deep learning (U-Net, MONAI, nnUNet) and image registration (Elastix).

Skin Lesion Classification

Oct 2024 – Jan 2025

- Developed a CAD pipeline combining classical image processing (ABCD, GLCM, Gabor filters) and deep learning (MIXNet, Swin Transformer, EfficientNet, ResNet) with weighted ensemble for improved diagnostic accuracy.

Multiclass & Semantic Segmentation of Colorectal Cancer Cells

Apr 2024 – July 2024

- Built early detection pipeline using multi-segmentation algorithms (82% Dice), feature extraction (LBP, Gabor) with machine learning (83% accuracy), and UNet+EfficientNet-B2 achieving 93% Dice.

FedCSIS 2024 Data Science Challenge

Mar 2024 – June 2024

- Achieved 22nd place among 77 teams by developing predictive models for trading actions using 8,000-instance financial datasets. Applied ML techniques for feature engineering, model selection, and hyperparameter tuning; conducted statistical analysis for trend detection; and ensured high-quality insights through data cleaning, preprocessing, and visualization.

IoT-based Smart Warehouse System

Aug 2021 – May 2022

- Designed and implemented a smart warehouse management system integrating IoT and RFID for real-time tracking, built with Node.js (backend) and ReactJS (frontend). Improved operational efficiency by enhancing inventory tracking and reducing errors, and optimized the prototype for deployment in real-world environments.